THE SHEAF WITH EXACT SUPPORT PRESERVES QUASI-COHERENT COMPLEXES AND AGREES WITH THE ALGEBRAIC HOM ON AFFINE CHARTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A78 (sources/stacks-duality/pinned-original-source.tex line 2373).

Allowed standard prerequisites (editorial).

- The sheaf with exact support and its computation. Same chapter, Lemma labelled lemma-sheaf-with-exact-support-ext, cited at proof lines 2400-2404 and 2438-2442, and Lemma labelled lemma-compute-sheaf-with-exact-support, cited at proof lines 2424-2428. Identifies RSheafHom(O_Z, -) with the object whose quasi-coherence is checked.
- Reflecting quasi-coherence along the pushforward of a closed immersion. Morphisms, Lemma labelled morphisms-lemma-i-star-equivalence, cited at proof lines 2398-2400. Reduces (1) to a statement about i_* of the cohomology sheaves.
- Internal Hom of a pseudo-coherent module stays quasi-coherent. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-internal-hom, cited at proof lines 2408-2412 and 2442-2446. Gives (1) and identifies the affine comparison map.
- Pseudo-coherence of a closed immersion into a locally Noetherian scheme. More on Morphisms, Lemma labelled more-morphisms-lemma-Noetherian-pseudo-coherent, cited at proof lines 2394-2396. Justifies the parenthetical remark in the statement.

Source reference. Original lemma 0684: sources/stacks-chow/tags.txt, source lines 7993–7993. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2395 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 01QY: sources/stacks-chow/tags.txt, source lines 2283–2283. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2399 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A77: sources/stacks-duality/pinned-original-source.tex, source lines 2339–2339. Same-chapter pointer into the author environment lemma at source line 2339; invoked at proof line 2403 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2410 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A75: sources/stacks-duality/pinned-original-source.tex, source lines 2283–2283. Same-chapter pointer into the author environment lemma at source line 2283; invoked at proof line 2424 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $i : Z \to X$ be a pseudo-coherent closed immersion of schemes (any closed immersion if X is locally Noetherian). Then*

(1) *$R\mathcal{H}om(\mathcal{O}_Z, -)$ maps $D^+_{QCoh}(\mathcal{O}_X)$ into $D^+_{QCoh}(\mathcal{O}_Z)$, and*
(2) *if $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(B)$, then the diagram*

$$\begin{array}{ccc} D^+(B) & \longrightarrow & D^+_{QCoh}(\mathcal{O}_Z) \\ {\scriptstyle R\mathrm{Hom}(B,-)}\uparrow & & \uparrow{\scriptstyle R\mathcal{H}om(\mathcal{O}_Z,-)} \\ D^+(A) & \longrightarrow & D^+_{QCoh}(\mathcal{O}_X) \end{array}$$

is commutative.

Proof. To explain the parenthetical remark, if X is locally Noetherian, then i is pseudo-coherent by More on Morphisms, Lemma 0684.

Let K be an object of $D^+_{QCoh}(\mathcal{O}_X)$. To prove (1), by Morphisms, Lemma 01QY it suffices to show that i_* applied to $H^n(R\mathcal{H}om(\mathcal{O}_Z, K))$ produces a quasi-coherent module on X. By Lemma 0A77 this means we have to show that $R\mathcal{H}om_{\mathcal{O}_X}(i_*\mathcal{O}_Z, K)$ is in $D_{QCoh}(\mathcal{O}_X)$. Since i is pseudo-coherent the sheaf $\mathcal{O}_Z$ is a pseudo-coherent $\mathcal{O}_X$-module. Hence the result follows from Derived Categories of Schemes, Lemma 0A6H.

Assume $X = \mathrm{Spec}(A)$ and $Z = \mathrm{Spec}(B)$ as in (2). Let $I^\bullet$ be a bounded below complex of injective A-modules representing an object K of $D^+(A)$. Then we know that $R\mathrm{Hom}(B, K) = \mathrm{Hom}_A(B, I^\bullet)$ viewed as a complex of B-modules. Choose a quasi-isomorphism

$$\widetilde{I^\bullet} \longrightarrow \mathcal{I}^\bullet$$

where $\mathcal{I}^\bullet$ is a bounded below complex of injective $\mathcal{O}_X$-modules. It follows from the description of the functor $\mathcal{H}om(\mathcal{O}_Z, -)$ in Lemma 0A75 that there is a map

$$\mathrm{Hom}_A(B, I^\bullet) \longrightarrow \Gamma(Z, \mathcal{H}om(\mathcal{O}_Z, \mathcal{I}^\bullet))$$

Observe that $\mathcal{H}om(\mathcal{O}_Z, \mathcal{I}^\bullet)$ represents $R\mathcal{H}om(\mathcal{O}_Z, \widetilde{K})$. Applying the universal property of the $\widetilde{\ }$ functor we obtain a map

$$\widetilde{\mathrm{Hom}_A(B, I^\bullet)} \longrightarrow R\mathcal{H}om(\mathcal{O}_Z, \widetilde{K})$$

in $D(\mathcal{O}_Z)$. We may check that this map is an isomorphism in $D(\mathcal{O}_Z)$ after applying i_*. However, once we apply i_* we obtain the isomorphism of Derived Categories of Schemes, Lemma 0A6H via the identification of Lemma 0A77. □